THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **Patterns in Life: Diversity and Classification** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Whale is placed in the class Mammalia. Reason (R): It has mammary glands and breathes with lungs. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A new organism is found to be multicellular, with cell walls of chitin, non-motile, and with thread-like hyphae. Its kingdom is: 1

(a) Animalia **(b)** Plantae **(c)** Fungi **(d)** Protista

3. Two animals A and B are in the same order. Which statement is certainly true? 1

(a) They are in the same genus
(b) They are in the same class and phylum
(c) They are the same species
(d) They are in the same family

4. A scientist finds an organism that has no cell wall, is multicellular, has a body with tissues, and feeds on other organisms. It is a(n): 1

(a) protist **(b)** animal **(c)** fungus **(d)** plant

5. Dolphins and sharks look similar but belong to different classes because: 1

(a) dolphins have gills
(b) dolphins have lungs, hair and mammary glands
(c) sharks have lungs
(d) both have fins

6. Assertion (A): Bryophytes are called the amphibians of the plant kingdom. Reason (R): They need water for the fertilisation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A student finds a plant with a vascular system, seeds in cones and no flowers. The class is: 1

(a) Angiosperms **(b)** Bryophytes **(c)** Pteridophytes **(d)** Gymnosperms

8. The binomial name 'Canis lupus' and 'Canis familiaris' show that the two: 1

(a) belong to different genera
(b) belong to the same genus but different species
(c) belong to the same species
(d) belong to different families

9. **Assertion (A):** The species is the basic unit of classification. 1
Reason (R): Members of a species can interbreed to produce fertile offspring.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The scientific name of the human being is *Homo sapiens*. 1
Reason (R): In binomial nomenclature the first word is the genus (beginning with a capital letter) and the second word is the species (beginning with a small letter).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do scientists keep revising the classification of organisms? Give two reasons. 2

12. Distinguish between homeothermic and poikilothermic animals with the classes. 2

13. Why do the chordates have a dorsal hollow nerve cord and invertebrates a ventral solid one? 2

14. Differentiate between Archaea and Bacteria in two points. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Construct a simple key (two choices at each step) to identify the following: pigeon, frog, shark, bat, snake. 3

16. Compare the life cycles of a moss and a fern briefly, noting the dominant stage in each. 3

17. State why some classification groups, such as algae, are not considered natural groups by modern taxonomists. 3

OR

A newly discovered organism is unicellular, has a cell wall, a nucleus, no chloroplasts and absorbs food from dead matter. Name its kingdom and give reasons. 3

18. Explain how classification reflects the evolutionary relationships using the example of the vertebrate classes. 3

OR

Explain why the amphibians had to evolve lungs and the reptiles needed a shelled egg for life on land. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A botany teacher in Lucknow writes the scientific names of three organisms on the board: mango, *Mangifera indica*; neem, *Azadirachta indica*; and tiger, *Panthera tigris*. 5
(i) Name the genus of the mango.
(ii) Name the species of the tiger.
(iii) Mango and neem both have the species name *indica*. Are they the same species?
(iv) Write the name of the tiger correctly according to the rules and give one reason why scientific names are preferred to common names.

20. Discuss how the major groups of animals show increasing complexity in body organisation, symmetry, coelom, segmentation, and the circulatory system with specific examples. 5

OR

A student notes the features of an animal: bilaterally symmetrical, body divided into head, thorax and abdomen, chitinous exoskeleton, a compound eye, six jointed legs and a pair of antennae. Identify the phylum and class, name three adaptations that help it survive and state its importance to humans. 5

END OF QUESTION PAPER — Total: 40 Marks